FEAT-ZD-48600 · DS-04 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Data Science agent · item task.components · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600 (Data Science lane, ap_forecast)

DS-04 Identify the components/layers to be changed (attempt 1). Workbook action: read the existing code, understand it, and list the components and layers to create or update with the minimum lines of code. Also confirm access to secrets in all environments.

Result: the lane changes one repository, ap_forecast. It needs one new source module (`risk.py`), additive edits to two existing modules (`explain.py`, `__init__.py`), one new test file (`tests/test_risk.py`) and appended tests in `tests/test_forecast.py`. `predict.py`, `models.json` and `ci.py` stay unchanged. No database, Databricks or secret is needed in any environment (section 5).

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-04 Identify the components/layers to be changed (attempt 1)	record:brief
Lane / repository	Data Science / ap_forecast	feat.tasks (section 1); task.repo_access (section 1)
Tasks in scope	TSK-03 (STORY-03), TSK-04 (STORY-04)	feat.tasks (sections 1, 3)
Inputs	feat.tasks (b0fd4d2), feat.stories (0590ea4), feat.prd (908b58d), feat.trd (3637e51), feat.design (722a98f), task.repo_access (aac590a)	record:brief
Background read	task.secrets_scope (DS-02, e779ba1), task.db_changes (DS-03, 27457ab)	journey:task.secrets_scope; journey:task.db_changes; mcp:hobasa-context (context_list revisions)
Code read	`src/ap_forecast/__init__.py`, `explain.py`, `predict.py`, `models.json`, `tests/test_forecast.py`; `ci.py`	tool:repo; file:ap_forecast/ci.py
Working branch	`feature/story`, created from `dev` at `d67e866`, clean	task.repo_access (section 2)

1. What the code does today

Layer	Module	Today	Source
Package API	`src/ap_forecast/__init__.py`	Imports `predict_delay_days` and `build_explanation_request`; `__version__ = "0.9.0"`; `__all__` lists those two names	tool:repo
Prediction (rule / model)	`src/ap_forecast/predict.py`	`predict_delay_days(history)`: rounded mean of past delays; 0 for an empty history; negative delays (early payments) count as 0	tool:repo

Layer	Module	Today	Source
LLM request	`src/ap_forecast/explain.py`	`build_explanation_request(vendor, predicted_days, config=None)` builds a one-sentence prompt and returns `provider`, `model`, `max_tokens` (from `config or load_config()`) and `messages`. It builds the dict; it does not send it	tool:repo
LLM configuration	`src/ap_forecast/models.json`	`provider` `acme-llm`, `model` `acme-text-2`, `max_tokens` 300; no key or endpoint	tool:repo
Tests	`tests/test_forecast.py`	`test_predict`, `test_request_uses_config`, `test_request_with_explicit_config`	tool:repo
CI	`ci.py`, run as `/usr/bin/python3 ci.py`	Compile, unittest discovery under `tests/`, coverage on `src/ap_forecast` with `COVERAGE_MIN = 90.0`, static checks (no `print` / `eval` / `exec`, no bare `except`, lines at most 140 characters) over every `src/ap_forecast/*.py`	file:ap_forecast/ci.py; task.repo_access (section 3)

The package has no network, pipeline or storage layer of its own (tool:repo; journey:task.db_changes, section 2). Because `ci.py` discovers every test file under `tests/` and measures coverage over every module in `src/ap_forecast`, a new `risk.py` and a new `tests/test_risk.py` are picked up without any CI change (file:ap_forecast/ci.py).

2. Components and layers to create or update

#	Layer	Component	Action	Task / story	What changes (minimum)	Source
CMP-1	Prediction (rule)	`src/ap_forecast/risk.py`: `HIGH_ABOVE_DAYS = 30`, `MEDIUM_ABOVE_DAYS = 10`, `vendor_risk(history: Iterable[int]) -> str`	**Create**	TSK-03 / STORY-03	New module. Imports only `predict_delay_days` (not `explain`). Calls it once and compares with strict >: `'high'` above 30, `'medium'` above 10, else `'low'`	feat.design (HC-4, B2.1, B2.2, DD-9); feat.trd (IF-3.1 to IF-3.3); feat.tasks (TSK-03); feat.prd (AC-7 to AC-10)
CMP-2	Package API	`src/ap_forecast/__init__.py`	**Update**	TSK-03 / STORY-03	Add `from .risk import vendor_risk`, add `"vendor_risk"` to `__all__`, and take the package's minor version step from `0.9.0` (TSK-03 owns it)	feat.design (B2.1, DD-11); feat.tasks (section 2, TSK-03); feat.stories (TK-03.1, TK-03.4, AC-03.8)

#	Layer	Component	Action	Task / story	What changes (minimum)	Source
CMP-3	LLM request	`src/ap_forecast/explain.py`: `build_explanation_request(vendor, predicted_days, config=None, risk=None)`	**Update (additive)**	TSK-04 / STORY-04	Add `risk=None` as the last parameter. Append `f" Risk level: {risk}."` to the prompt only when `risk is not None`. The returned dict and `load_config` stay the same	feat.design (HC-5, B2.2); feat.trd (IF-4.1 to IF-4.5); feat.tasks (TSK-04); feat.prd (AC-11, AC-12)
CMP-4	Tests	`tests/test_risk.py`	**Create**	TSK-03 / STORY-03	`test_vendor_risk_levels`, `test_vendor_risk_boundaries`, `test_vendor_risk_empty_and_early`, `test_vendor_risk_no_llm`, `test_vendor_risk_exported`	feat.tasks (section 2 test files, TSK-03); feat.design (B2.5)
CMP-5	Tests	`tests/test_forecast.py`	**Update (append)**	TSK-04 / STORY-04	Add `test_request_with_risk` and `test_request_without_risk_unchanged`. Keep the existing three tests exactly as they are	feat.tasks (TSK-04); feat.design (B2.5); feat.stories (TK-04.2)

Left unchanged

Component	Why	Source
`src/ap_forecast/predict.py`	`vendor_risk` reuses `predict_delay_days` as it is	feat.design (B2.1); feat.tasks (TSK-03 "Must not change")
`src/ap_forecast/models.json`	Provider, model and `max_tokens` are unchanged; `vendor_risk` must not read it	feat.trd (IF-3.2, IF-4.4); feat.tasks (TSK-03, TSK-04 "Must not change")
`load_config` and the `config or load_config()` source in `explain.py`	Provider, model and `max_tokens` still come from `config` or `models.json`	feat.stories (AC-04.4)
Existing signatures	"Keep every existing signature; add new functions." `risk` goes last so positional `config` calls still work	feat.design (Architecture rule row, NF-4); feat.prd (NFR-4); feat.stories (AC-04.3)
`ci.py`, `tests/__init__.py`, `README.md`, `CLAUDE.md`	Nothing in TSK-03 or TSK-04 needs them changed; CI already discovers new tests and modules	file:ap_forecast/ci.py; feat.tasks (TSK-03, TSK-04 "Files in budget")
Other repositories (ap_pipeline, ap_aging, ap_portal)	Other lanes' tasks. No lane imports another lane's package	feat.design (Part C "Lane dependencies"); task.repo_access (section 1)

3. Minimum-code implementation sketch

The sketches follow the signatures and rules in feat.design B2.2. DS-05 writes the final code.

CMP-1, new `risk.py`. A single comparison chain with no validation of its own: `predict_delay_days` already applies `int(d)`, and calling it once lets a generator work (feat.tasks TSK-03 notes).

```
"""Rule-based vendor payment-risk level from the predicted delay. No LLM, no models.json read."""
from typing import Iterable

from .predict import predict_delay_days

HIGH_ABOVE_DAYS = 30
MEDIUM_ABOVE_DAYS = 10


def vendor_risk(history: Iterable[int]) -> str:
    """'high' if the predicted delay is above HIGH_ABOVE_DAYS, 'medium' if above MEDIUM_ABOVE_DAYS, else 'low'."""
    days = predict_delay_days(history)
    if days > HIGH_ABOVE_DAYS:
        return "high"
    return "medium" if days > MEDIUM_ABOVE_DAYS else "low"
```

CMP-2, `__init__.py`. One import line, one name added to `__all__`, and the `__version__` value changed.

CMP-3, `explain.py`. Add the parameter and one guarded append. Everything else in the function stays the same.

```
def build_explanation_request(vendor: str, predicted_days: int, config=None, risk=None) -> dict:
    cfg = config or load_config()
    prompt = f"Explain in one sentence why vendor {vendor} is predicted to be paid {predicted_days} days late."
    if risk is not None:
        prompt += f" Risk level: {risk}."
    return {...}  # unchanged
```

Why this is the minimum

Choice	Effect on code size	Source
Reuse `predict_delay_days` instead of a second averaging rule	No change to `predict.py`	feat.design (B2.1); feat.trd (IF-3.1)
No input validation in `vendor_risk`; call `predict_delay_days` exactly once	No extra branches to test	feat.tasks (TSK-03 notes)
`risk` appended without validation (any non-`None` value)	One `if` in `explain.py`	feat.tasks (TSK-04 notes); feat.trd (IF-4.5)
`risk` as the last parameter with default `None`	No caller or existing test changes	feat.trd (IF-4.1); feat.stories (AC-04.3)
Export only `vendor_risk` in `__all__`; the threshold constants stay module-level in `risk.py` (agent judgement)	No extra exports to maintain	feat.design (B2.1); feat.stories (TK-03.1, AC-03.8)
No CI or config change	`ci.py` already covers new modules and test files	file:ap_forecast/ci.py

4. Tests and coverage

Test	File	Asserts	AC	Source
`test_vendor_risk_levels`	`tests/test_risk.py`	Histories predicting 31 → `'high'`, 20 → `'medium'`, 5 → `'low'`	AC-03.1, AC-03.3	feat.design (B2.5)
`test_vendor_risk_boundaries`	`tests/test_risk.py`	Single-value histories `[30]` → `'medium'`, `[31]` → `'high'`, `[10]` → `'low'`, `[11]` → `'medium'`	AC-03.2, AC-03.4	feat.tasks (TSK-03); record:memory (L-007)
`test_vendor_risk_empty_and_early`	`tests/test_risk.py`	`[]` → `'low'`; only negative delays → `'low'`	AC-03.5, AC-03.6	feat.design (B2.5)

Test	File	Asserts	AC	Source
`test_vendor_risk_no_llm`	`tests/test_risk.py`	Patch `ap_forecast.explain.load_config` to fail; `vendor_risk` still returns	AC-03.7	feat.tasks (TSK-03)
`test_vendor_risk_exported`	`tests/test_risk.py`	`"vendor_risk" in ap_forecast.__all__`	AC-03.8	feat.design (B2.5)
`test_request_with_risk`	`tests/test_forecast.py`	Prompt ends with `'Risk level: high.'`	AC-04.1	feat.tasks (TSK-04)
`test_request_without_risk_unchanged`	`tests/test_forecast.py`	Same result with no `risk` and with explicit `risk=None`	AC-04.2, AC-04.5	feat.tasks (TSK-04)
Existing three tests	`tests/test_forecast.py`	Pass unchanged (positional `config` still works)	AC-04.3	feat.design (B2.5)

Agent judgement from the sketches in section 3: every new line is reached by these tests, because each branch of `vendor_risk` is hit by the levels and boundaries tests and both sides of the `risk is not None` guard are hit by the two request tests, so the coverage bar of at least 90% (feat.tasks section 2 CI bar; file:ap_forecast/ci.py) is expected to hold. CI has not run in this lane yet, and whether the `coverage` package is installed on the runner is still open (task.repo_access RAQ-1); the first CI run in DS-05/DS-06 confirms both.

5. Secrets, database and environments

Item	Finding	Source
Secrets added by this change	None. "No secrets, credentials or personal data are added"	feat.trd (S-6); feat.design (SE-4)
Credentials in ap_forecast code or config	None. `models.json` holds only `provider`, `model` and `max_tokens`	tool:repo
Outbound calls added	None. `vendor_risk` makes no LLM or network call; `build_explanation_request` only builds the request dict, with provider and model unchanged	feat.trd (IF-3.2, IF-4.4); tool:repo
Secrets per environment (CI, `feature/<story>`, `dev`, `main`)	None required and none granted in any environment; no access request needed	journey:task.secrets_scope (section 3)
Database / Databricks, all environments	None needed; nothing to provision	journey:task.db_changes (section 3); task.repo_access (section 2)

The tests need no live provider: they check the built prompt text and patch `load_config` (feat.tasks TSK-03, TSK-04). Sending the risk label to `acme-llm` waits on Architect confirmation (TQ-8), which gates release, not coding (feat.tasks TSK-04; feat.design SE-3). Whichever component eventually sends the request will need a provider credential, outside this lane (journey:task.secrets_scope SSQ-1).

6. Dependencies and order

Item	Effect on this lane	Source
Other lanes	No code dependency. TSK-03 and TSK-04 can be built in parallel with DE, BE and FE	feat.design (Part C); feat.tasks (section 1)
TSK-03 ↔ TSK-04	Independent. They touch different files apart from the shared version step, which TSK-03 owns	feat.tasks (section 2)
PROV-ACME-2026-11 (PQ-5)	Also changes `build_explanation_request`. Whichever merges second rebases and re-runs T-7	feat.tasks (TSK-04); feat.trd (R-3); feat.prd (PQ-5)
End-to-end harness (TSK-08)	Uses `vendor_risk` (CMP-1); does not call `build_explanation_request`	feat.tasks (TSK-08); feat.stories (SA-5)

7. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
CQ-DS-1	feat.stories TK-03.2 suggests adding the risk tests to `tests/test_forecast.py`, while feat.tasks (section 2 "Test files"; TA-4 in section 6) assigns a new `tests/test_risk.py`	Follow feat.tasks, which is the later task breakdown: new `tests/test_risk.py`; the split still awaits Developer confirmation (TKQ-2)	Developer
CQ-DS-2	Exact next version number after `0.9.0`	A minor step, value confirmed in DS-05 (feat.tasks section 2: "The next version number is for the lane to confirm")	Developer
RAQ-1 (carried)	Is `coverage` installed for `/usr/bin/python3` on the runner?	Assume yes; first CI run confirms	Developer / infra
RAQ-2 (carried)	Both stories on the single `feature/story` branch, or per-story branches?	Shared `feature/story`, commits prefixed `STORY-03` / `STORY-04`	Developer

Carried, unchanged: PQ-1 (AC-03.5), TQ-5 (AC-03.6), TQ-4 (AC-04.5), TQ-8 and PQ-5 (TSK-04 release) (feat.stories; feat.tasks); SSQ-1 (journey:task.secrets_scope).

8. Validation check

Check	Result	Source
Every DS task maps to components	TSK-03 → CMP-1, CMP-2, CMP-4; TSK-04 → CMP-3, CMP-5	feat.tasks (TSK-03, TSK-04)
Files stay inside each task's budget	Yes	feat.tasks (TSK-03, TSK-04 "Files in budget")
Existing signatures kept	Yes (`risk=None` is the last parameter)	feat.design (NF-4); feat.trd (IF-4.1)
No new technology	Python 3.9 standard library only	feat.design (DD-1)
Decisions and lessons	L-007 applied (boundary tests `[30]`, `[31]`, `[10]`, `[11]`). Consistent with D-2026-044 / L-011 (feat.tasks section 8 lists TSK-03 as consistent); the ap_forecast code handles delay days and labels, not monetary amounts (agent judgement from the code)	record:memory; feat.tasks (TSK-03, section 8); tool:repo
Conflict with an approved output	None; the test-file difference is listed as CQ-DS-1	feat.stories (TK-03.2); feat.tasks (section 2)
Blocking questions	None	—